

Exploratory Data Analysis

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Dataset choice

For this project, we will be using the Airline Service Quality Performance 234 dataset from the Bureau of Transportation Statistics. The data was reported by each carrier as mandated by the Bureau of Transportation Statistics. The purpose is to make performance data available to airline consumers.

A collection of this data can be found at <https://www.bts.gov/browse-statistical-products-and-data/bts-publications/airline-service-quality-performance-234-time>.

This project will use a year's worth of United Airlines data between October 2024 and September 2025.

Data set fields description

Outcomes:

- **Departure.Delay** (continuous) = Difference between scheduled departure time and actual departure time. A summary of this column is:

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## -66.00   -7.00   -3.00   12.44    5.00 2081.00
```

- **Arrival.Delay** (continuous) = Difference between scheduled arrival time and actual arrival time. A summary of this column is:

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## -84.000  -17.000   -7.000    8.332    9.000 2152.000
```

Predictors:

- **Marketing.ft.num** (Categorical) = Four digit number used by United Airlines to identify a market. Some values include:

```
## [1] 5004 4976 4974 4970 4963 4962
```

There are a total of

```
## [1] 4518
```

values.

- **Operating.Airline** (Categorical) = Airline that operates the flight. This includes United (UA) and their regional partners. Operating.Airline takes the values:

```
## [1] "C5" "G7" "00" "UA" "YV" "YX"
```

- **Operating.flt.num** (Categorical) = Four digit number used by the operating airline to identify a market. Some values include:

```
## [1] 5004 4976 4974 4970 4963 4962
```

There are a total of

```
## [1] 4544
```

values.

- **Dep** (Categorical) = Three letter IATA code that defines the departure airport. Some values include:

```
## [1] "IAH" "LRD" "ALB" "IAD" "GSP" "SHV"
```

There are a total of

```
## [1] 227
```

values.

- **Arr** (Categorical) = Three letter IATA code that defines the arrival airport. Some values include:

```
## [1] "GPT" "LBB" "IAH" "IAD" "GSP" "SHV"
```

There are a total of

```
## [1] 227
```

values.

- **Report_Month** (Categorical) = Month the flight is scheduled in, taking the values:

```
## [1] 1 2 3 4 5 6 7 8 9 10 11 12
```

- **Day.of.Week** (Categorical) = Day of the week the flight is scheduled on, taking the values:

```
## [1] "Mon" "Tue" "Wed" "Thu" "Fri" "Sat" "Sun"
```

- **Scheduled.Departure.Time** (Continuous) = Minutes since midnight when a flight was scheduled to depart the gate. A summary of this column is:

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      2.0   552.0   793.0   801.3  1043.0  1439.0
```

- **Scheduled.Arrival.Time** (Continuous) = Minutes since midnight when a flight was scheduled to arrive at the gate. A summary of this column is:

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1.0   678.0   925.0   914.3  1147.0  1439.0
```

- **Scheduled.Block.Time** (Continuous) = Amount of time the flight is schedule to take, including taxi out and taxi in time. A summary of this column is:

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      30.0   95.0   130.0   149.6   180.0   687.0
```

- **Tail.Num** (Categorical) = Unique identifier for the aircraft. Some values include:

```
## [1] "N11193" "N12135" "N11189" "N16183" "N14174" "N31131"
```

There are a total of

```
## [1] 1563
```

values.

CoNVO statement

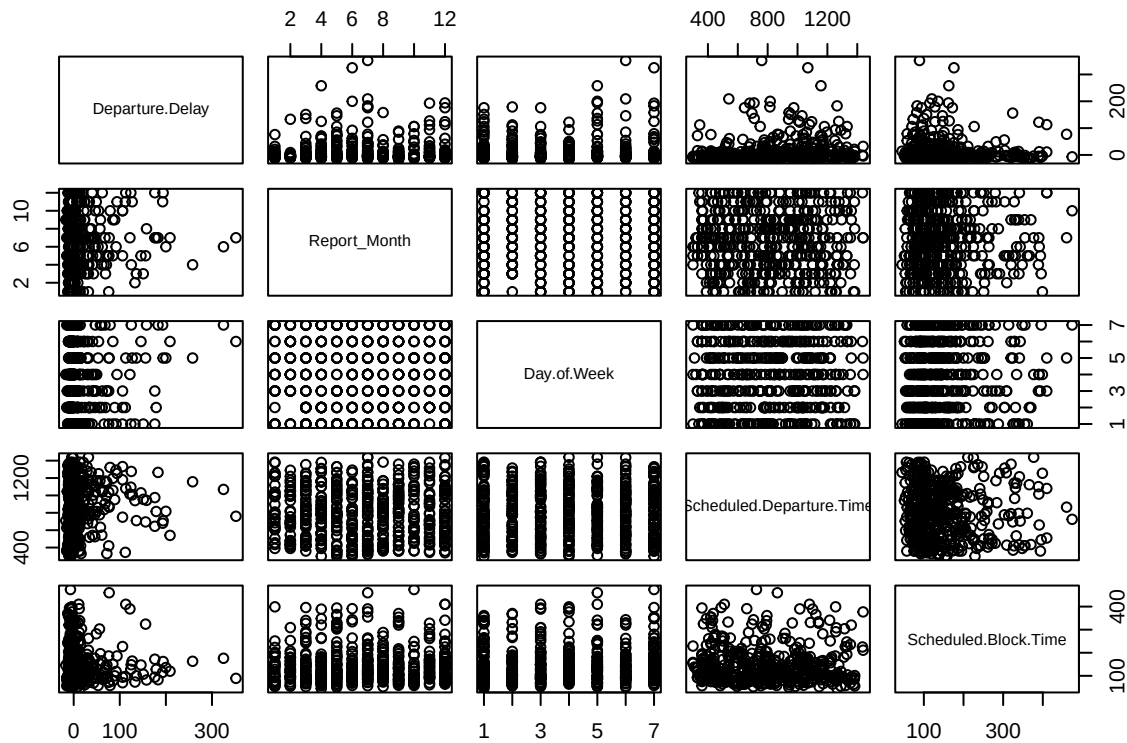
United Airlines operates thousands of flights every day across a large and complex network, and identifying flights at high risk of delay months in advance is a key challenge for the network planning team. Having better visibility into which flights are likely to be delayed is an important step toward building more reliable and operationally efficient schedules.

Understanding what drives flight delays will improve foresight into where and when delays are most likely to occur. This will help highlight routes and markets that are consistently delayed, giving planning teams a clearer picture of where the network is underperforming. With this information, better long-term scheduling decisions can be made that improve on-time performance and reduce delays have across the network.

Using historical flight data, we will build a model that predicts both the probability and intensity of departure delays for flights across the network. By identifying the key factors that contribute to delays, the model will give network planning teams the information they need to adjust routes and time slots where schedule changes could improve reliability and operational performance.

Preliminary results will be presented to a VP in Network Planning. If successful, the model will be run across all flights in the network on an ongoing basis, giving planning teams regular insights into where delays are expected and where schedules may need to be adjusted. The long term goal is to make this model a regular part of the network planning process to help the team build better schedules and reduce delays across the network.

Pairs plot

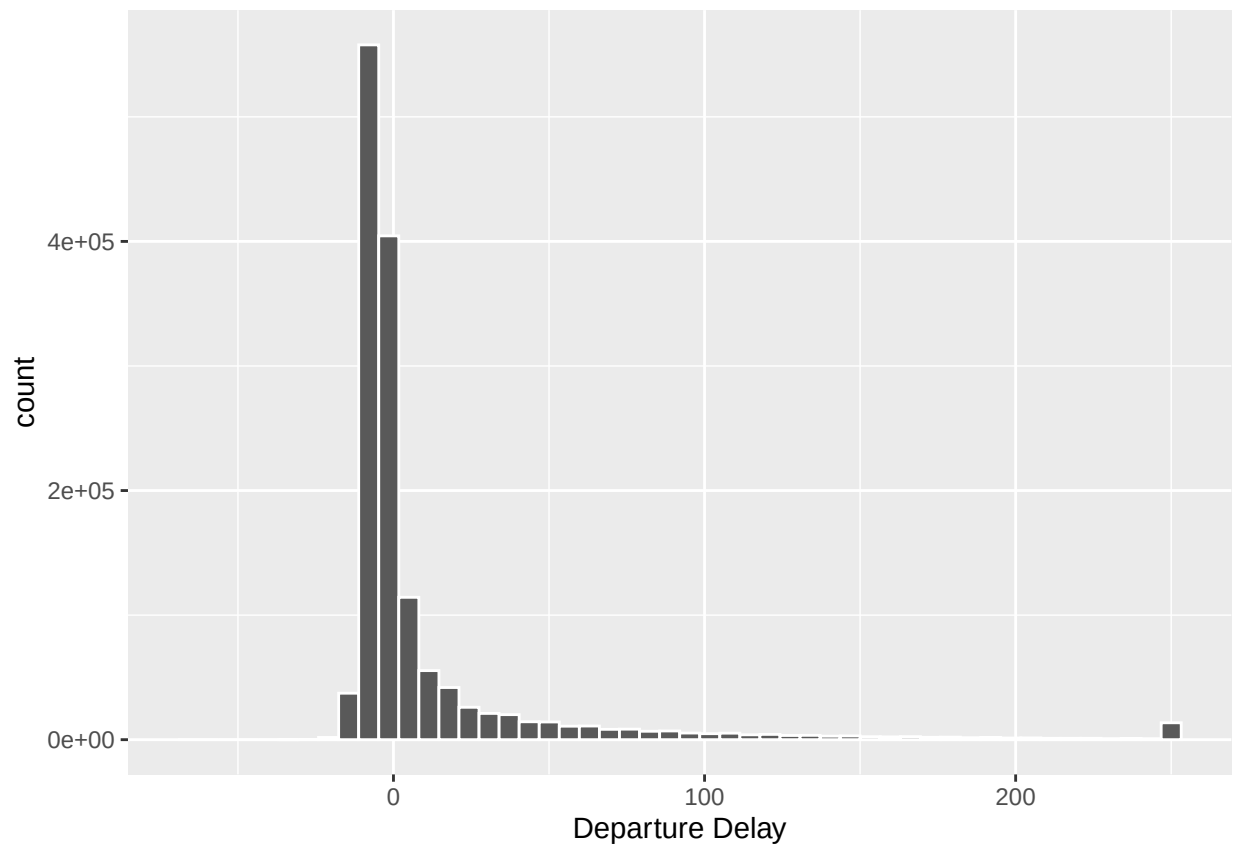


Identify an outcome and three predictor variables

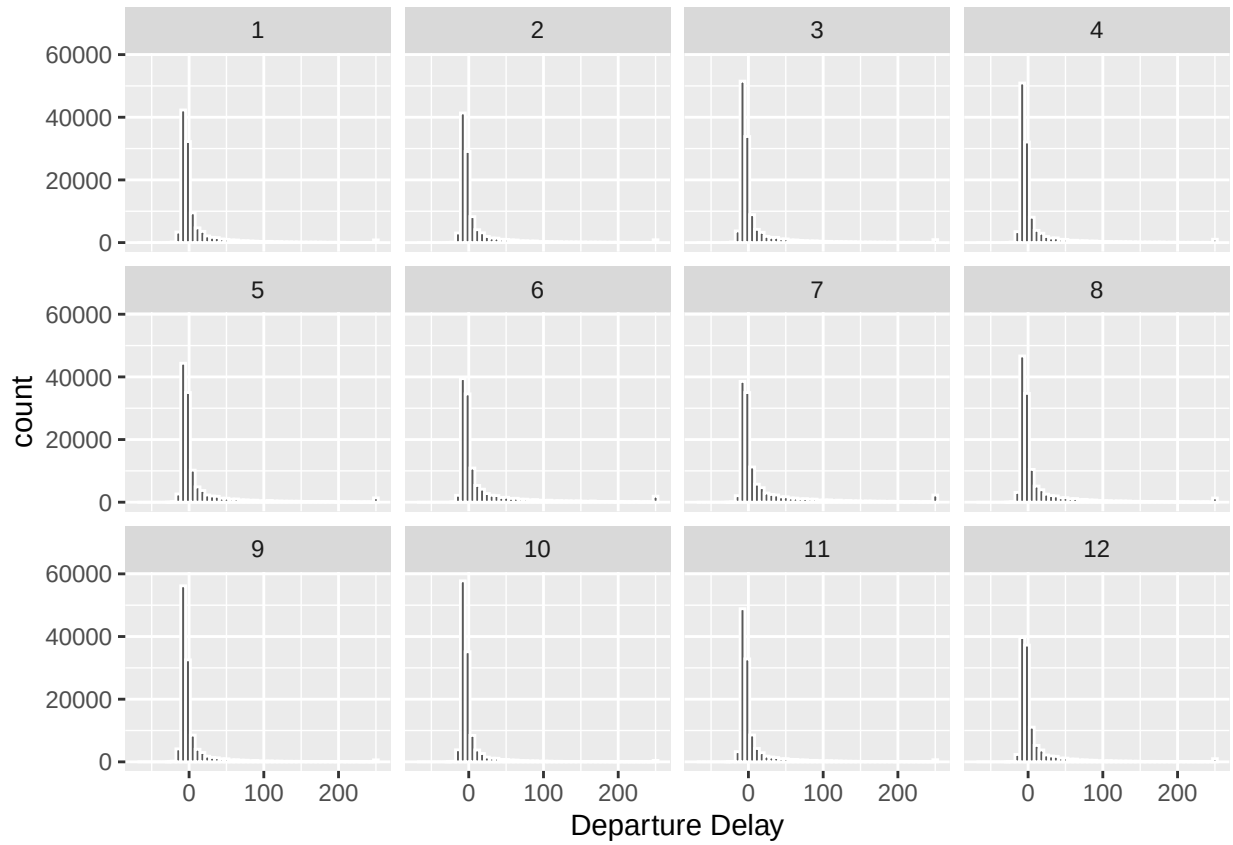
The outcome variable that was selected is Departure.Delay. Another possible variable that could be looked at is Arrival.Delay, though the predictors in the dataset are better applied to departure delays.

The predictor variables that were selected are Report_Month, Day.of.Week and Scheduled.Departure.Time as they appear to be uncorrelated but each have a relationship with the outcome Departure.Delay.

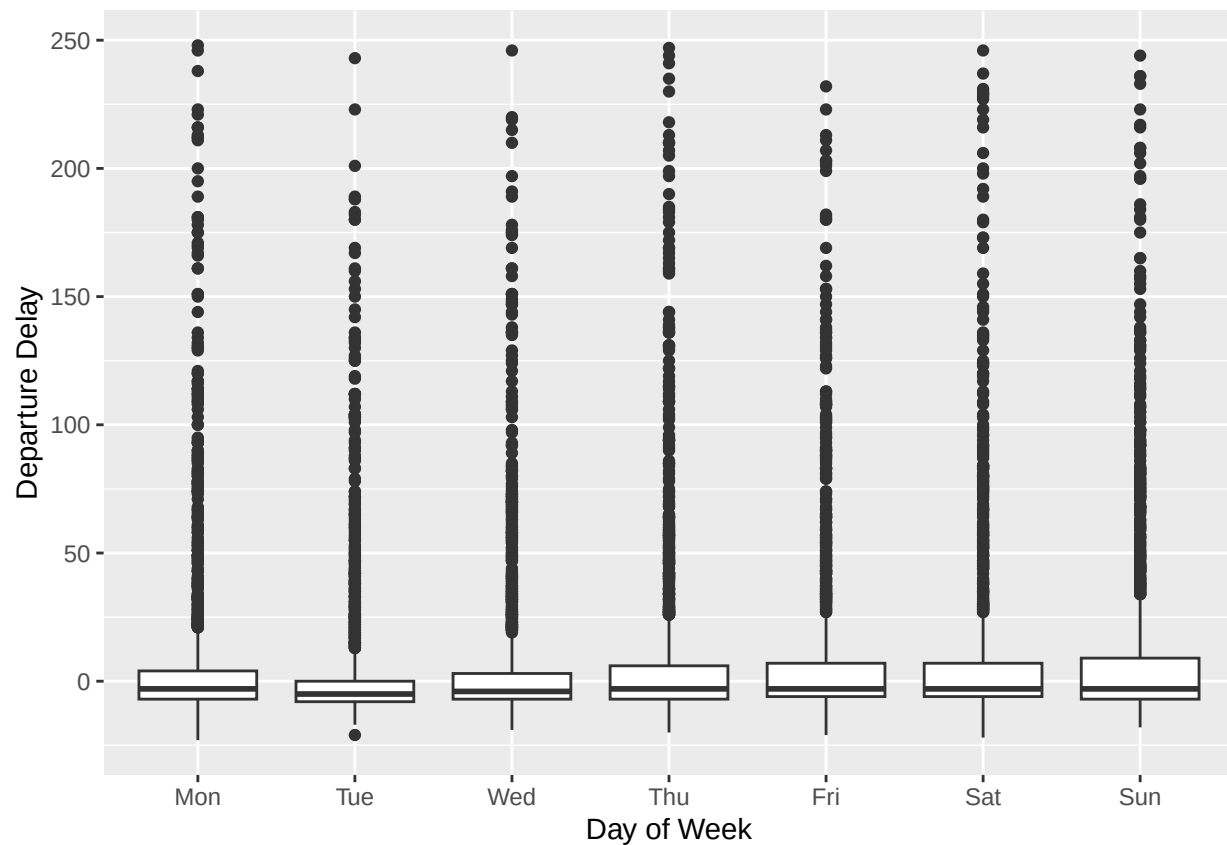
Explore the variables by applying four plot types



The histogram above shows how departure delays are distributed. This distribution is heavily skewed to the right, as 75% of the delay values are less than 5 minutes. Note: The max value `Departure.Delay` took was 2081. The tail is much longer than visualized but is grouped into a bucket at 250 minutes.

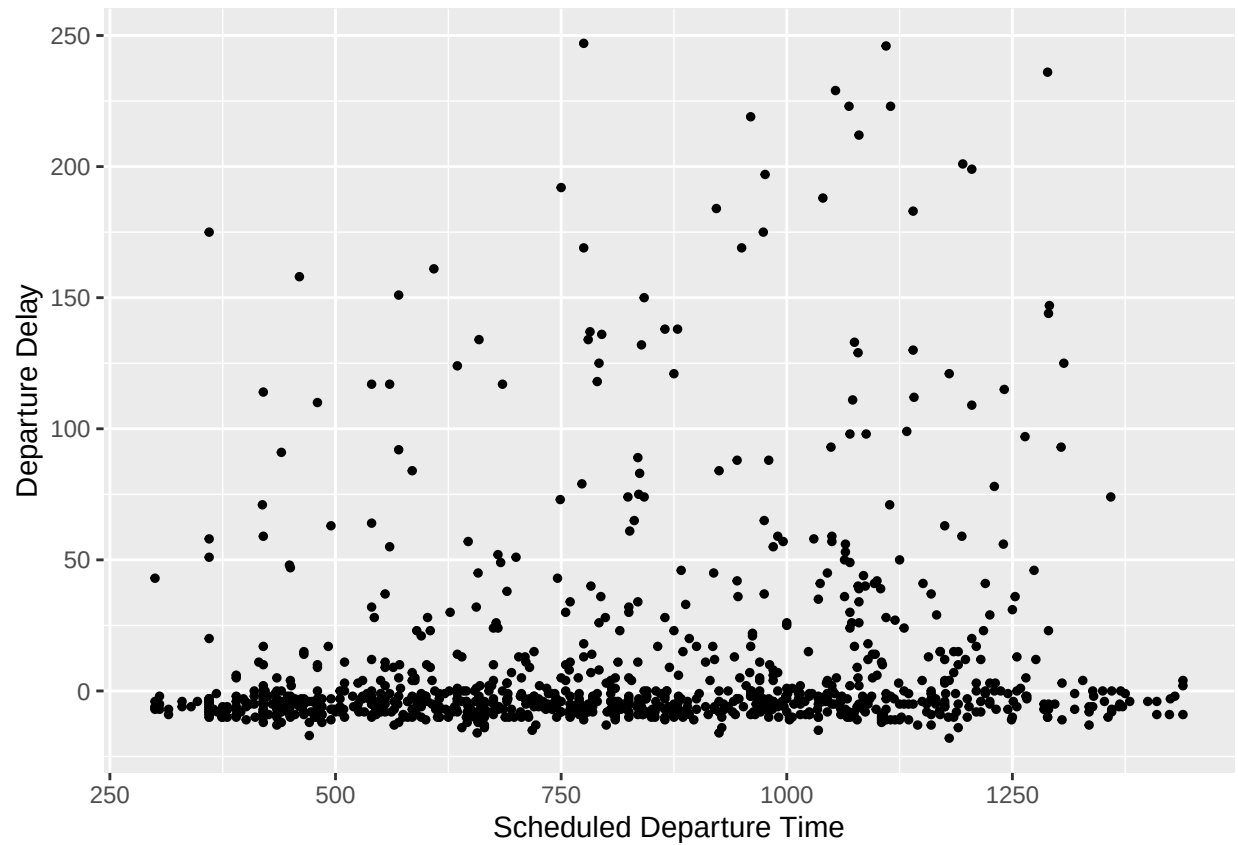


The above facet grid of histograms shows how Departure Delays are distributed each month. September and October have significantly more flights departing on time compared to the summer months.



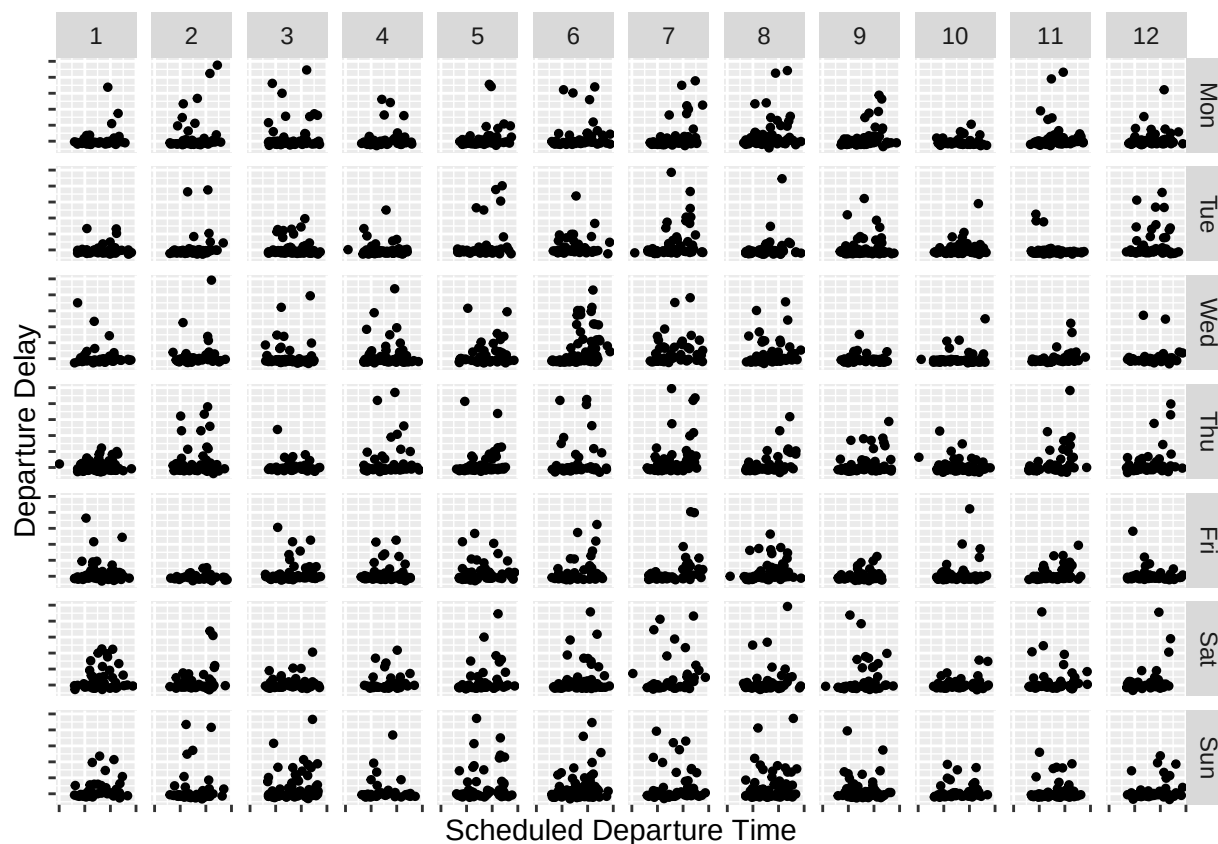
```
## # A tibble: 7 x 3
##   Day.of.Week    p75    p95
##   <fct>         <dbl> <dbl>
## 1 Mon             6    92
## 2 Tue             1  62.8
## 3 Wed             3  67.4
## 4 Thu             8  94.4
## 5 Fri             7   73
## 6 Sat             7   80
## 7 Sun             6  74.3
```

The box plots and table above show differences in percentiles between each day of the week. Tuesday and Wednesday have the smallest 75th percentiles (ex: 75% of Tuesday flights depart within 1 minutes of scheduled), with Thursday through Monday having larger 75th percentiles.



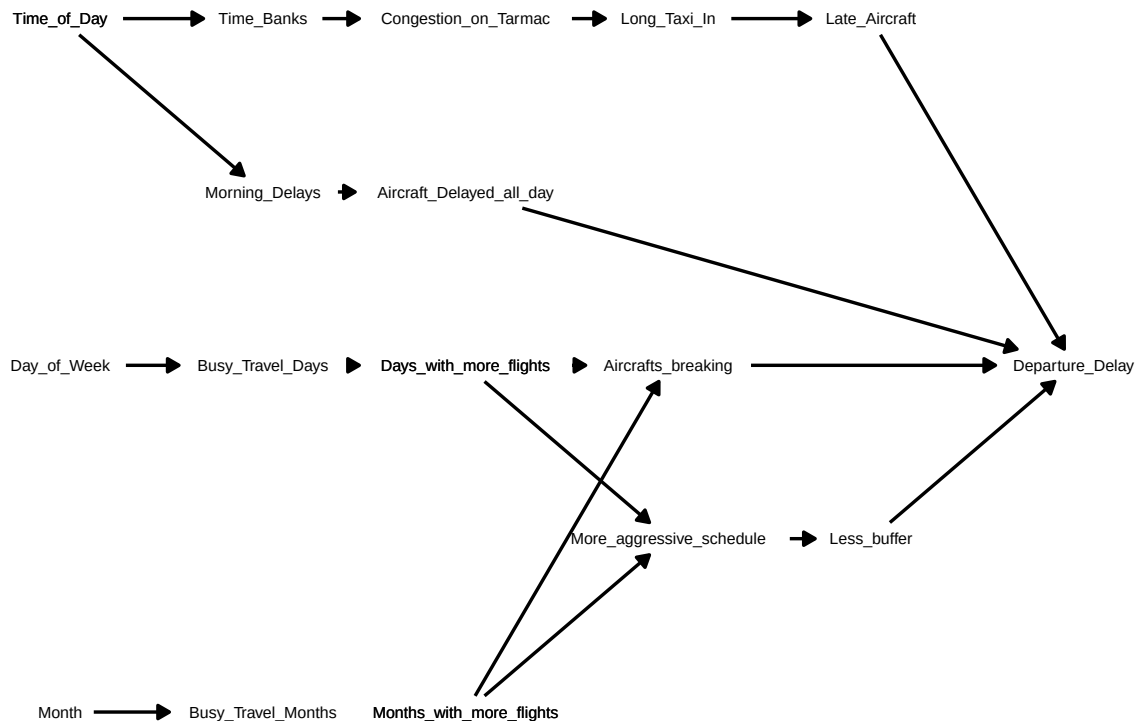
The above scatterplot shows the relationship between departure delay and scheduled departure time. While it isn't a strong relationship in this sample, there seems to be more long delays later in the day.

Single plot that shows the effects of all three predictors



The plot above shows how all 3 variables combine to effect departure delays. While this plot is somewhat overwhelming to look at, it may be helpful in identifying trends on a daily level during certain times of year. For example, Wednesday afternoons during June and Saturday midday during January have a larger concentration of long delays than most other combinations. A view like this could be especially helpful when looking at the schedule of an individual airport to identify times of heavy congestion that may be impacting departure delays.

Causal relationships



The causal diagram above shows some of the possible causal relationships between the variables. For both month and week, there are naturally days where flying is more in demand. Airlines will provide more flights during these periods, which results in a higher maintenance rate and a more aggressive schedule which includes less buffer between flights to make up lost time earlier in the day. When looking at time of day, cascading delays (flights delayed early in the day resulting in late aircraft and delays later in the day) as well as busy times on the tarmac (United has a “bank” structure which contributes at hubs, though traffic from all airlines causes this) result in aircraft arriving late at the gate which cause the next outbound flight to depart late as well.

Difficulties preparing the data

Getting the data all in one table was difficult. Since the datasets were so large and came in asc files, they needed to be imported into excel by each month and then combined. There were too many rows for an excel sheet so they had to be combined using R and saved to a csv. Several variables needed to be converted to factors and month had to be manually calculated from the date. Times were also saved as hhmm, so all time variables had to be converted to minutes after 00:00.

Improved analysis

To improve this analysis, I would like to combine this with a weather dataset. This would present a whole new set of factors that impact delays. In further phases of the project, I will attempt to create new variables from the current dataset. For example, determining the taxi-in time during the previous hour at the station may be significant with departure delay with a confounding variable of congestion on the ramp at the airport.